

# PE-EC801B — Fiber Optic Communication

Semester VIII • 3L:0T:0P • 3 Credits

## Module 1 Exam Notes

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## 1. Vector Nature of Light — EM Wave Fundamentals

### Light as an Electromagnetic Wave

Light is a **transverse electromagnetic (TEM) wave** in which the electric field  $\vec{E}$  and magnetic field  $\vec{H}$  are perpendicular to each other and to the direction of propagation  $\hat{k}$ . In free space, both fields satisfy Maxwell's wave equation.

### 1.1 Maxwell's Equations and Wave Equation

For a linear, isotropic, source-free dielectric medium (permittivity  $\epsilon$ , permeability  $\mu$ ), Maxwell's curl equations are:

$$\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}, \quad \nabla \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t}$$

Taking the curl of the first equation and using the second gives the vector wave equation:

$$\nabla^2 \vec{E} = \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

The speed of light in the medium is  $v = 1/\sqrt{\mu\epsilon}$ . In free space,  $v = c = 3 \times 10^8$  m/s, where  $c = 1/\sqrt{\mu_0\epsilon_0}$ .

### 1.2 E and H Field Relationship

For a plane wave propagating in the  $+z$  direction with wavenumber  $k = \omega\sqrt{\mu\epsilon}$ :

$$\vec{E} = E_0 e^{j(\omega t - kz)} \hat{x}, \quad \vec{H} = \frac{E_0}{\eta} e^{j(\omega t - kz)} \hat{y}$$

where  $\eta = \sqrt{\mu/\epsilon}$  is the intrinsic impedance of the medium. In free space,  $\eta_0 = 377 \Omega$ .

### Key EM Wave Parameters

|                     |                                                                                     |
|---------------------|-------------------------------------------------------------------------------------|
| Wave number         | $k = \frac{2\pi}{\lambda} = \frac{\omega}{v} = \omega\sqrt{\mu\epsilon}$            |
| Refractive index    | $n = \frac{c}{v} = \sqrt{\mu_r\epsilon_r} \approx \sqrt{\epsilon_r}$ (non-magnetic) |
| Intrinsic impedance | $\eta = \sqrt{\mu/\epsilon}$ ; free space $\eta_0 = 377 \Omega$                     |
| Poynting vector     | $\vec{S} = \vec{E} \times \vec{H}$ (power flow direction)                           |

## 2. Propagation of Light — Reflection, Refraction, and TIR

### Snell's Law and Critical Angle

When light travels from medium 1 (refractive index  $n_1$ ) to medium 2 ( $n_2$ ), **Snell's law** governs the angle of refraction:  $n_1 \sin \theta_1 = n_2 \sin \theta_2$ . The **critical angle**  $\theta_c$  is the angle of incidence in medium 1 at which the refracted ray in medium 2 travels along the interface ( $\theta_2 = 90^\circ$ ). **Total Internal Reflection (TIR)** occurs for  $\theta_1 > \theta_c$  when  $n_1 > n_2$ .

### Snell's Law and Critical Angle Formulas

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad (\text{Snell's Law})$$

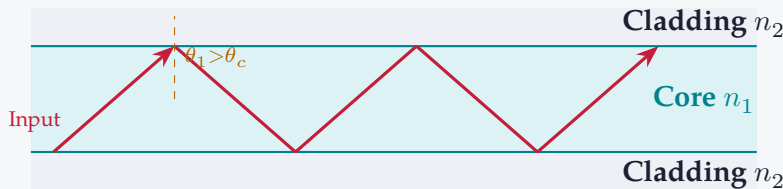
$$\sin \theta_c = \frac{n_2}{n_1} \quad (n_1 > n_2) \Rightarrow \theta_c = \sin^{-1} \left( \frac{n_2}{n_1} \right)$$

TIR condition:  $\theta_1 \geq \theta_c$  (light stays in the denser medium).

## 2.1 Propagation in a Cylindrical Dielectric Rod (Optical Fiber)

An optical fiber is a cylindrical dielectric rod comprising a **core** (refractive index  $n_1$ ) surrounded by a **cladding** (refractive index  $n_2 < n_1$ ). Light injected at the core-cladding interface at an angle  $\geq \theta_c$  undergoes TIR repeatedly and is guided along the fiber without escaping.

### TIR in an Optical Fiber — Ray Path



## 3. Ray Model — Acceptance Angle and Numerical Aperture

### Numerical Aperture (NA) and Acceptance Angle

The **acceptance angle**  $\theta_a$  (or half-cone angle) is the maximum angle at which light may enter the fiber end-face and still undergo TIR. The **Numerical Aperture (NA)** is a dimensionless measure of the light-gathering ability of the fiber:  $NA = n_0 \sin \theta_a$  where  $n_0$  is the refractive index of the medium at the input (usually air,  $n_0 = 1$ ).

### 3.1 Derivation of NA

Consider a ray entering the flat end of the fiber at angle  $\theta_i$  (in air). By Snell's law at the end-face:

$$\sin \theta_i = n_1 \sin \theta_r$$

where  $\theta_r$  is the refraction angle inside the core. For TIR at the core–cladding interface, the ray must strike that interface at an angle  $\geq \theta_c$ , meaning the complementary angle  $\theta_r \leq (90^\circ - \theta_c)$ . Using  $\sin \theta_c = n_2/n_1$ :

#### Numerical Aperture — Complete Formula

$$\text{NA} = \sin \theta_a = \sqrt{n_1^2 - n_2^2}$$

$$\text{NA} = n_1 \sqrt{2\Delta}, \quad \Delta = \frac{n_1^2 - n_2^2}{2n_1^2} \approx \frac{n_1 - n_2}{n_1} \quad (\text{relative refractive index difference})$$

Acceptance angle:  $\theta_a = \sin^{-1}(\text{NA})$  (full acceptance cone =  $2\theta_a$ ).

### 3.2 Meridional vs Skew Rays

- **Meridional rays:** Rays that pass through the fiber axis. Their path lies entirely in a plane containing the axis. They follow simple zigzag trajectories and are the easiest to analyze.
- **Skew rays:** Rays that do not pass through the fiber axis. They travel in a helical path around the axis. They have a larger acceptance angle than meridional rays and carry higher-order modes.

#### Worked Example: Calculate NA and Acceptance Angle

Given:  $n_1 = 1.48$ ,  $n_2 = 1.46$ . Find NA and  $\theta_a$ .

$$\text{NA} = \sqrt{1.48^2 - 1.46^2} = \sqrt{2.1904 - 2.1316} = \sqrt{0.0588} \approx 0.2425$$

$$\theta_a = \sin^{-1}(0.2425) \approx 14.03^\circ, \quad \Delta = \frac{1.48 - 1.46}{1.48} \approx 1.35\%$$

## 4. Wave Model — Modes and V-Number

### Modes in an Optical Fiber

A **mode** is a distinct spatial distribution (pattern) of the electromagnetic field that can propagate along the fiber without changing shape. Only certain discrete field patterns satisfy the boundary conditions at the core–cladding interface; these are the allowed or guided modes.

#### 4.1 Normalized Frequency (V-Number)

The **V-number** (also called the normalized frequency or normalized waveguide parameter) determines how many modes a fiber can support.

#### V-Number Formula

$$V = \frac{2\pi a}{\lambda} \text{NA} = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi a}{\lambda} n_1 \sqrt{2\Delta}$$

where  $a$  = core radius,  $\lambda$  = free-space wavelength of light.

- $V < 2.405 \Rightarrow$  **single-mode** fiber (only the  $HE_{11}$  fundamental mode propagates).
- $V \geq 2.405 \Rightarrow$  **multimode** fiber.
- Total number of modes (step-index):  $M \approx \frac{V^2}{2}$ .
- Total number of modes (graded-index, parabolic):  $M \approx \frac{V^2}{4}$ .

#### 4.2 Cutoff Condition and Mode Designation

Modes are designated  $HE_{mn}$  or  $EH_{mn}$  (hybrid modes),  $TE_{0n}$ ,  $TM_{0n}$ . The fundamental mode  $HE_{11}$  has no cutoff. The first higher-order mode  $TE_{01}$  cuts off at  $V = 2.405$  (first zero of Bessel function  $J_0$ ).

#### 4.3 Modal Analysis of a Step-Index Fiber

The electromagnetic field inside the core satisfies the wave equation in cylindrical coordinates, yielding Bessel functions  $J_n$  (oscillatory, inside core) and modified Bessel functions  $K_n$  (decaying, inside cladding). Applying continuity of tangential  $E$  and  $H$  at the core-cladding boundary gives the **characteristic equation** (eigenvalue equation) for the allowed propagation constants  $\beta$ :

##### Characteristic Equation for Step-Index Fiber

$$\left[ \frac{J'_n(U)}{U J_n(U)} + \frac{K'_n(W)}{W K_n(W)} \right] \left[ \frac{n_1^2 J'_n(U)}{U J_n(U)} + \frac{n_2^2 K'_n(W)}{W K_n(W)} \right] = n^2 \left( \frac{1}{U^2} + \frac{1}{W^2} \right)^2$$

where the normalized transverse wave numbers are:

$$U = a \sqrt{n_1^2 k_0^2 - \beta^2}, \quad W = a \sqrt{\beta^2 - n_2^2 k_0^2}, \quad U^2 + W^2 = V^2$$

For the  **$LP_{01}$  ( $HE_{11}$ ) fundamental mode**: the simplified characteristic equation is  $J_0(U)K_1(W) + J_1(U)K_0(W) = 0$ . This transcendental equation is solved numerically to find  $\beta$  for each mode.

#### Key results from modal analysis:

- Each solution  $\beta_m$  corresponds to one guided mode with phase velocity  $v_{ph} = \omega/\beta_m$ .
- Group velocity:  $v_g = d\omega/d\beta$  (determines pulse propagation and intermodal dispersion).
- For LP modes:  $LP_{lm}$  modes correspond to  $HE_{l+1,m}$  and  $EH_{l-1,m}$  modes. The LP approximation ( $n_1 \approx n_2$ ) simplifies the characteristic equation to  $J_{l-1}(U)/[U J_l(U)] = -K_{l-1}(W)/[W K_l(W)]$ .

#### Worked Example: V-Number and Mode Count

Given:  $a = 25 \mu\text{m}$ ,  $\lambda = 1300 \text{ nm}$ ,  $n_1 = 1.48$ ,  $n_2 = 1.46$ .

$$\text{NA} = 0.2425 \quad (\text{from previous example})$$

$$V = \frac{2\pi \times 25 \times 10^{-6}}{1300 \times 10^{-9}} \times 0.2425 = \frac{2\pi \times 25}{1.3} \times 0.2425 \approx 29.2$$

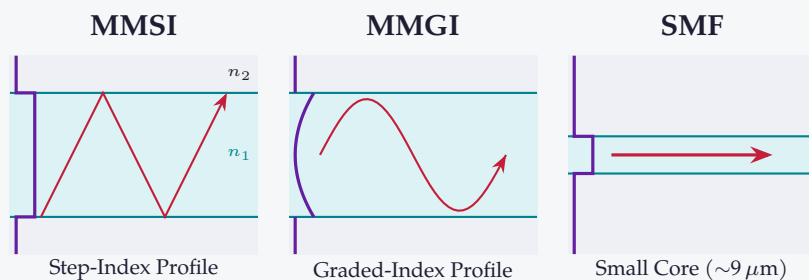
Number of modes  $M \approx V^2/2 \approx 426$ . Since  $V \gg 2.405$ , this is a multimode fiber.

## 5. Types of Optical Fibers

### Classification of Optical Fibers

Optical fibers are classified by (a) **index profile** of the core — step-index (SI) or graded-index (GI), and (b) **number of modes** — single-mode (SM) or multimode (MM). The three main types are: Single-Mode Step-Index (SMF), Multimode Step-Index (MMSI), and Multimode Graded-Index (MMGI).

### Refractive Index Profiles and Ray Paths — Three Fiber Types



### 5.1 Comparison Table — SMF vs MMSI vs MMGI

| Parameter           | SMF                                              | MMSI                                                      | MMGI                                                            |
|---------------------|--------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------|
| Core diameter       | $\sim 8\text{--}10\ \mu\text{m}$                 | $\sim 50\text{--}200\ \mu\text{m}$                        | $\sim 50\text{--}62.5\ \mu\text{m}$                             |
| Cladding diameter   | $125\ \mu\text{m}$                               | $125\text{--}400\ \mu\text{m}$                            | $125\ \mu\text{m}$                                              |
| Index profile       | Step                                             | Step                                                      | Parabolic ( $\alpha \approx 2$ )                                |
| V-number            | $< 2.405$                                        | $\gg 2.405$                                               | $\gg 2.405$                                                     |
| No. of modes        | 1 ( $\text{HE}_{11}$ only)                       | $\sim V^2/2$ (hundreds)                                   | $\sim V^2/4$                                                    |
| Bandwidth           | Very high<br>( $>10\ \text{GHz}\cdot\text{km}$ ) | Low<br>( $\sim 10\text{--}25\ \text{MHz}\cdot\text{km}$ ) | Medium<br>( $\sim 400\text{--}2000\ \text{MHz}\cdot\text{km}$ ) |
| Modal dispersion    | None                                             | High                                                      | Low (near-zero for $\alpha=2$ )                                 |
| NA                  | $\sim 0.1\text{--}0.12$                          | $\sim 0.2\text{--}0.5$                                    | $\sim 0.2\text{--}0.3$                                          |
| Coupling efficiency | Low (small core)                                 | High                                                      | Medium                                                          |
| Light source        | Laser only                                       | LED or Laser                                              | LED or Laser                                                    |
| Application         | Long-haul, WDM                                   | Short-haul, LAN                                           | LAN, medium distance                                            |

## Quick Revision — Module 1

Light Propagation and Optical Fiber Fundamentals | PE-EC801B

### Key Formulas at a Glance

|                             |                                                             |
|-----------------------------|-------------------------------------------------------------|
| <b>Snell's Law</b>          | $n_1 \sin \theta_1 = n_2 \sin \theta_2$                     |
| <b>Critical angle</b>       | $\theta_c = \sin^{-1}(n_2/n_1) \quad (n_1 > n_2)$           |
| <b>Numerical Aperture</b>   | $NA = \sqrt{n_1^2 - n_2^2} = n_1 \sqrt{2\Delta}$            |
| <b>Relative index diff.</b> | $\Delta = (n_1 - n_2)/n_1$ (approx. for $n_1 \approx n_2$ ) |
| <b>V-number</b>             | $V = (2\pi a/\lambda) NA$                                   |
| <b>SMF condition</b>        | $V < 2.405$                                                 |
| <b>Mode count (MMSI)</b>    | $M \approx V^2/2$                                           |
| <b>Mode count (MMGI)</b>    | $M \approx V^2/4$                                           |
| <b>Wave speed in medium</b> | $v = c/n; \quad n = \sqrt{\epsilon_r}$                      |

### One-Line Definitions

- **TIR:** Total Internal Reflection — light stays in denser medium when  $\theta_i \geq \theta_c$ .
- **NA:** Numerical Aperture — measure of light-gathering ability of the fiber.
- **V-number:** Normalized frequency parameter determining number of guided modes.
- **Meridional ray:** Ray passing through the fiber axis, zigzag path.
- **Skew ray:** Ray traveling in a helical path, does not pass through axis.
- **Mode:** Distinct stable field distribution pattern guided along the fiber.
- **SMF:** Single-mode fiber with  $V < 2.405$ ; eliminates modal dispersion.
- **MMGI:** Multimode graded-index fiber; parabolic profile minimizes modal dispersion.

### Frequently Asked / Viva Questions

#### Q1. What is Total Internal Reflection and when does it occur?

TIR occurs when light travels from a denser medium ( $n_1$ ) to a rarer medium ( $n_2 < n_1$ ) and the angle of incidence exceeds the critical angle  $\theta_c = \sin^{-1}(n_2/n_1)$ . All incident light is reflected back; no energy enters medium 2.

#### Q2. Why is the refractive index of the core greater than the cladding in an optical fiber?

TIR — which is the guiding mechanism — requires light to travel from a denser

## Frequently Asked / Viva Questions (contd.)

to rarer medium. Thus  $n_1 > n_2$  is essential for guided propagation.

**Q3. Define Numerical Aperture (NA). What does a higher NA imply?**

$NA = \sqrt{n_1^2 - n_2^2}$  is the sine of the maximum acceptance angle. Higher NA means the fiber accepts light over a wider cone but also supports more modes (greater modal dispersion).

**Q4. What is the V-number and what is its significance?**

The V-number  $= (2\pi a/\lambda) NA$  is the normalized frequency. It determines the number of guided modes. For  $V < 2.405$ , only one mode propagates (single-mode operation).

**Q5. Distinguish between meridional and skew rays.**

Meridional rays pass through the fiber axis and travel in a plane containing it. Skew rays do not intersect the axis; they travel helically. Skew rays have a slightly larger effective acceptance angle.

**Q6. Why does a graded-index fiber have lower modal dispersion than step-index?**

In MMGI, rays in the outer (faster) region travel longer paths but at higher speeds (lower  $n$ ), while rays near the axis travel slower but shorter paths. This self-equalizing effect reduces pulse spreading.

**Q7. What is the cutoff condition for single-mode operation?**

$V < 2.405$ , where 2.405 is the first zero of the Bessel function  $J_0$ . Below this, only the fundamental  $HE_{11}$  mode propagates.

**Q8. What is  $\Delta$  (relative refractive index difference)?**

$\Delta = (n_1 - n_2)/n_1 \approx (n_1^2 - n_2^2)/(2n_1^2)$ . It is typically 0.1–2% for communication fibers. Larger  $\Delta$  means higher NA but also higher dispersion.

**Q9. How does the number of modes depend on V-number?**

For step-index:  $M \approx V^2/2$ . For graded-index (parabolic):  $M \approx V^2/4$ . The graded profile halves the mode count for the same V-number.

**Q10. What are the dimensions of a standard SMF and MMSI fiber?**

SMF: core  $\sim 8\text{--}10\ \mu\text{m}$ , cladding  $125\ \mu\text{m}$ . MMSI: core  $50\text{--}200\ \mu\text{m}$ , cladding  $125\ \mu\text{m}$ . MMGI: core  $50\text{--}62.5\ \mu\text{m}$ , cladding  $125\ \mu\text{m}$ .

**Q11. State the wave equation for the electric field in a dielectric medium.**

$\nabla^2 \vec{E} = \mu\epsilon \partial^2 \vec{E} / \partial t^2$ . The speed of propagation is  $v = 1/\sqrt{\mu\epsilon} = c/n$ .

**Q12. What is the Poynting vector and what does it represent in fiber optics?**

$\vec{S} = \vec{E} \times \vec{H}$  ( $\text{W/m}^2$ ). It represents the direction and magnitude of electromagnetic power flow. In a fiber, it points along the fiber axis for guided modes.

## Summary: Fiber Type Comparison at a Glance

| Feature    | SMF                | MMSI                 | MMGI                  |
|------------|--------------------|----------------------|-----------------------|
| V-number   | $< 2.405$          | Large                | Large                 |
| Core dia.  | 8–10 $\mu\text{m}$ | 50–200 $\mu\text{m}$ | 50–62.5 $\mu\text{m}$ |
| Dispersion | Lowest             | Highest              | Low                   |
| Bandwidth  | Highest            | Lowest               | Medium                |
| Source     | Laser              | LED/Laser            | LED/Laser             |
| Use case   | Long haul          | Short links          | LAN/campus            |